

Cauê Adriano de Souza Castro

Software Engineer

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

Professional Summary

Software Engineer with hands-on experience across the full lifecycle of web, mobile, desktop, and system-integrated applications. Author of **Quick Lofi**, a GNOME extension with over **10,000 downloads** on the official store. Strong proficiency in **React**, **Next.js**, **TypeScript**, **Node.js**, **Python**, and **React Native**, building scalable REST APIs, responsive interfaces, and cross-platform solutions. Leverages AI-assisted development tools such as GitHub Copilot, Claude, and Gemini to accelerate delivery, automate workflows, and improve code quality.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Python, Shell Script, GJS

Frontend: React, Next.js, Vite, TailwindCSS, Shadcn/ui, HTML5, CSS3

Backend: Node.js, Express.js, Elysia, NestJS, Django, Django REST Framework (DRF)

Mobile: React Native, Expo

Desktop: GTK4, LibAdwaita, GJS, GObject, GLib

Databases: PostgreSQL, MongoDB, SQLite, Supabase

DevOps & Tools: Docker, GitHub Actions, CI/CD, Git, Linux, Meson, Flatpak, Vercel

Testing: Vitest, Unit Testing, Integration Testing

Methodologies: Agile, Scrum, Kanban, Code Review, REST API Design, Responsive Design

AI & Productivity: GitHub Copilot, Claude Code, OpenCode, ChatGPT, Gemini, Codex, DeepSeek, MCP (Model Context Protocol)

Professional Experience

Software Engineer (Junior) | VXCASE | Jan 2026 - Present

Mission Summary: Developing and maintaining a complex corporate ecosystem, with a primary focus on evolving a robust enterprise ERP. My work directly impacts internal operations and business workflows, ensuring efficient integration across services, high data availability, and continuous delivery of new features driven by code quality, scalability, and AI-enhanced productivity throughout the development lifecycle.

How we build this in practice:

- **Architecture & Backend:** Building scalable and resilient REST APIs using **Express**, **Elysia**, and **TypeScript**, enforcing clear architectural patterns (Route, Schema, Service, Repository) and

ensuring consistent contracts with **Zod** validations.

- **Frontend & Interface:** Developing dynamic, responsive interfaces and reusable components for web platforms and the ERP using **React**, **Next.js**, and **TailwindCSS**, delivering optimized performance and usability.
- **Data Engineering & Integration:** Modeling, querying, and managing data across relational and non-relational databases, while orchestrating communication between frontend, backend, internal microservices, and third-party systems.
- **Quality, DevOps & Agile Culture:** Ensuring application stability through automated testing with **Vitest**, continuous performance optimization, **CI/CD** pipelines with **Docker**, assertive code reviews, and AI tools to automate tasks and streamline workflows managed in **ClickUp**.

Full Stack Developer (Volunteer) | Associação Capacitare | Apr 2026 - Present

Mission Summary: Serving as a volunteer Full Stack Developer building the MVP of a digital platform focused on democratizing access to essential services (health, legal, and more) provided by volunteer professionals. My work directly impacts the end-to-end delivery of system interfaces and user flows, enabling a scalable connection between professionals and people in vulnerable situations.

How we build this in practice:

- **Frontend & Interface:** Developing complete and responsive system screens with a Mobile-first approach using **React**, **Vite**, and **Tailwind CSS**, ensuring high fidelity to **Figma** prototypes and accessibility standards.
- **Backend & Data Engineering:** Implementing business logic, secure authentication, and data communication using **Supabase (BaaS)** and **PostgreSQL**, guaranteeing data persistence, integrity (RLS), and scheduling rules for the MVP.
- **Agile Culture & Product:** Actively participating in task refinement and structured execution based on a **Kanban** workflow, collaborating with the product team to translate the PRD (Product Requirements Document) into high-value deliverables using **Trello** and **Notion**.
- **DevOps & Quality:** Ensuring consistent deliverables through rigorous version control with **Git**, functional validation of complex user journeys, and continuous production deployment using **Vercel**.

Frontend Developer (Volunteer) | Central Institute of Citizenship (ICEC) | 2025

Mission Summary: Led the conception and implementation of a digital platform focused on online fundraising for the institute. The project modernized the donation process by creating a direct, automated, and accessible financial flow that reduced friction for donors and expanded the organization's social impact.

How we build this in practice:

- **Product Engineering & Payments:** Developed an optimized donation page with native integration to the **PIX** payment system, focusing on security and reducing barriers for the end

user.

- **Backend Logic & Integration:** Implemented a dynamic **PIX QR Code** generation engine, processing complex payloads that embed real-time value and location data.
- **Sustainable Architecture:** Structured the codebase and content management with project longevity in mind, empowering non-technical NGO staff to perform basic maintenance and updates without constant reliance on developers.

Projects

Quick Lofi | GNOME Extension

[GitHub](#) | [Preview](#)

Mission Summary: Native extension for the GNOME desktop environment focused on providing a fast, integrated audio experience. The project reached over 10,000 downloads on the official store, delivering an optimized solution that enhances user workflow and focus through a minimalist interface.

How we build this in practice:

- **Operating System Integration:** Low-level integration with the Linux desktop environment using **GJS** (GNOME JavaScript bindings) for native UI manipulation.
- **Media Engineering & IPC:** Asynchronous audio playback engine based on **mpv**, managing remote communication and control via **IPC** (Inter-Process Communication).
- **Frontend & UX:** User interface seamlessly integrated into the system status bar, prioritizing quick access, low latency, and visual minimalism.

Flexa | Cross-platform Desktop Application

[GitHub](#)

Mission Summary: Cross-platform desktop application aimed at automating operating system customization, running natively on both Linux and Windows. The project solves a persistent pain point in the customization community by automating bidirectional cursor theme conversion between Windows and Linux distributions.

How we build this in practice:

- **Cross-platform Desktop Development:** Entire graphical interface and presentation logic built with **Python**, **GTK4**, and **LibAdwaita**, ensuring compatibility and native feel across different operating systems.
- **Asynchronous Processing & Concurrency:** Robust execution architecture using **threads** and **queues** to manage heavy, bidirectional file conversion tasks without blocking the main UI thread.
- **Automation Engineering:** Automated conversion scripts and logic to transcribe Windows pointer assets into Linux technical specifications, as well as the reverse flow.

- **DevOps, CI/CD & Packaging:** Project build structured with **Meson**, artifact generation automated through **CI/CD pipelines with GitHub Actions**, and cross-platform packaging including **Flatpak** and Windows installers.

Auto Volume | Mobile Application | React Native

[GitHub](#)

Mission Summary: Mobile utility app focused on hardware automation (audio control) aimed at preventing sudden loud sounds and hearing damage when connecting headphones. The application runs invisibly and ensures the device predictably and safely returns to configured volume levels.

How we build this in practice:

- **Mobile Development:** Core application and settings interface built using the **React Native (Expo)** framework.
- **Native Integration & Events:** Continuous system listeners that intercept and monitor state changes of physical volume buttons in real-time.
- **Background Processing:** Continuous and asynchronous execution with advanced timer management, allowing the automation to run natively in the **background** while keeping the user informed through persistent OS notifications.

Scrolled | Firefox Extension

[GitHub](#)

Mission Summary: Lightweight extension for the Firefox browser aimed at improving the reading experience and accessibility on long articles and documentation by adding precise visual reading progress indicators.

How we build this in practice:

- **Frontend Development:** Scroll detection logic and viewport calculations with responsive DOM manipulation using **TypeScript**.
- **Performance Optimization:** Build pipeline structured with **Rollup**, ensuring an extremely lean final bundle that does not impact the loading performance of web pages.

Education

Associate Degree in Systems Analysis and Development

UniAmérica | Completed in 2025

Certifications & Languages

- Advanced English (C2) | EF SET
- Python AI Backend Developer | DIO
- MongoDB Data Modeling and CRUD Operations | MongoDB

- Object-Oriented Programming
- DevOps Strategy